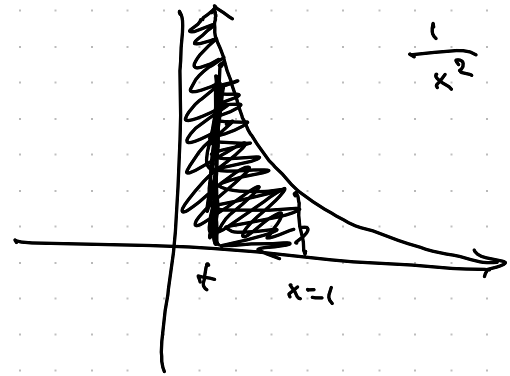
Improper integrals (type II) — 3.7



Discontinuous at $x=0$

Definition: The following are improper integrals (type II)

- f is continuous on $[a,b)$

$$\int_a^b f(x)dx = \lim_{t\to b^-} \int_a^t f(x)dx$$

- f is continuous on $(a,b]$

$$\int_a^b f(x)dx = \lim_{t\to a^+} \int_t^b f(x)dx$$

- f is continuous on $[a,b]$ except at $x=c$

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

<u>Example</u>: Does $\int_0^1 \frac{1}{x^2}\,dx$ converge or diverge?

$$= \lim_{t\to 0^+} \int_t^1 \frac{1}{x^2}\,dx = \lim_{t\to 0^+} -\frac{1}{x}\Big|_t^1 = \lim_{t\to 0^+} -\frac{1}{1} + \frac{1}{t}$$

$$= -1 + \lim_{t\to 0^+} \frac{1}{t} = \infty$$

So $\int_0^1 \frac{1}{x^2}\,dx$ diverges, but $\int_1^\infty \frac{1}{x^2}\,dx$ converges

(previous video)

Example: Does $\int_0^2 x\ln(x)\,dx$ converge or diverge?

$$= \lim_{t\to 0^+} \int_t^2 x\ln(x)\,dx \qquad u = \ln(x) \quad dv = x \quad \Rightarrow \quad du = \frac{1}{x} \quad v = \frac{x^2}{2}$$

$$= \lim_{t\to 0^+} \left[\frac{x^2}{2}\ln(x)\Big|_t^2 - \int_t^2 \frac{x}{2}\,dx \right]$$

$$= \lim_{t\to 0^+} \left[\frac{x^2}{2}\ln(x)\Big|_t^2 - \frac{x^2}{4}\Big|_t^2 \right]$$

$$= \lim_{t\to 0^+} \left[2\ln(2) - 1 - \frac{t^2}{2}\ln(t) + \frac{t^2}{4} \right]$$

$$= 2\ln(2) - 1 + \cancelto{0}{\lim_{t\to 0^+} \frac{t^2}{4}} - \cancelto{0}{\lim_{t\to 0^+} \frac{t^2}{2}\ln(t)}$$

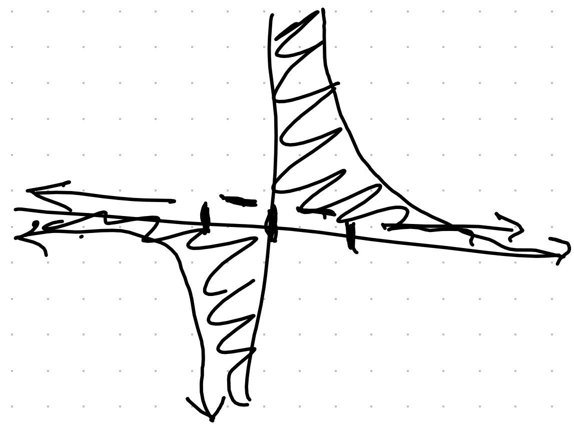
$$= 2\ln(2) - 1, \text{ so } \int_0^2 x\ln(x)\,dx \text{ converges.}$$

$$\lim_{t\to 0^+} \frac{\overset{\to -\infty}{\ln(t)}}{\underset{\to \infty}{\frac{1}{t^2}}} \overset{L'H}{=} \lim_{t\to 0^+} \frac{\frac{1}{t}}{\frac{-2}{t^3}} = \lim_{t\to 0^+} \frac{1}{t}\cdot\frac{-t^3}{2} = \lim_{t\to 0^+} \frac{-t^2}{2} = 0$$

Type III improper integrals are integrals that are both type I and type II improper.

<u>Example</u>: $\displaystyle\int_{-\infty}^{\infty} \frac{1}{x^3}\,dx$ is type III improper

(and it diverges)



$$\int_{-\infty}^{\infty} \frac{1}{x^3}\,dx = \lim_{M\to-\infty} \int_M^0 \frac{1}{x^3}\,dx + \lim_{N\to\infty} \int_0^N \frac{1}{x^3}\,dx$$

$$= \underbrace{\lim_{M\to-\infty} \int_M^{-1} \frac{1}{x^3}\,dx}_{\text{converges } \checkmark} + \underbrace{\lim_{t\to0^-} \int_{-1}^{t} \frac{1}{x^3}\,dx}_{\text{diverges } \times} + \underbrace{\lim_{s\to0^+} \int_s^{1} \frac{1}{x^3}\,dx}_{\text{diverges } \times} + \underbrace{\lim_{N\to\infty} \int_1^{N} \frac{1}{x^3}\,dx}_{\text{converges } \checkmark}$$